

Chinese Remainder Theorem

I have always been interested to write an article about the Chinese Remainder Theorem. First it is because of its name, but more importantly it is because of its many applications. In short, of course the theorem provides us a mean to handle large numbers, using smaller numbers. In the West, it was usually called the Chinese Remainder Theorem. But in China, it has been referred as the theorem of Sun-Tzu. It is interesting to understand how the theorem was developed. But perhaps it is more interesting to see how it is used. For more practical applications, search for “Chinese Remainder Theorem” in Widedpedia.

Theorem (Chinese Remainder Theorem). Given a system of congruences to different moduli:

$$\begin{aligned}x &\equiv a_1 \pmod{m_1} \\x &\equiv a_2 \pmod{m_2} \\&\dots \\x &\equiv a_r \pmod{m_r}.\end{aligned}$$

Suppose that each pair of moduli is relatively prime: $\gcd(m_i, m_j) = 1$ for $i \neq j$.

Then there exists a simultaneous solution x to all the congruences, and any two solutions are congruent to one another modulo $M = m_1 m_2 \dots m_r$.

Proof. (Uniqueness) If x and x' are two solutions, then $x - x'$ is congruent to 0 modulo each of m_i , hence to M .

(Existence) Now let $M_i = \frac{M}{m_i}$, then $\gcd(m_i, M_i) = 1$, hence we can find N_i, P_i

such that $M_i N_i + m_i P_i = 1$, or $M_i N_i \equiv 1 \pmod{m_i}$. Take $x = \sum_{i=1}^r a_i M_i N_i$ and we are done.

Remark: Indeed we may formulate the theorem in a ring-theoretic situation.

(Chinese Remainder Theorem). Let I_1 and I_2 be ideals of a ring R which is *relatively prime* in the sense that $I_1 + I_2 = R$. If a_1 and a_2 are elements of R , then there exist an $a \in R$ such that $a \equiv a_i \pmod{I_i}$. Hence we have ring isomorphism: $R / (I_1 \cap I_2) \cong R / I_1 \oplus R / I_2$. The case can be generalized to more than two ideals.

“Proof”. From the ring R to the direct sum of quotient rings $R / I_1 \oplus R / I_2$, by sending x to $(x + I_1, x + I_2)$, we clearly have a homomorphism. The hard part is to

show that it is an epimorphism (surjective). Indeed let $(x_1 + I_1, x_2 + I_2)$, we want to find x such that $x + I_1 = x_1 + I_1$ and $x + I_2 = x_2 + I_2$. Now since the two ideals are relatively prime, we may choose $a_1 \in I_1$ and $a_2 \in I_2$, such that $a_1 + a_2 = 1$. Now set $x = a_1 x_2 + a_2 x_1$ (note the indices). Then

$$\begin{aligned} x - x_1 &= x_2 a_1 + x_1 a_2 - x_1 (a_1 + a_2) = (x_2 - x_1) a_1 \in I_1; \\ x - x_2 &= x_2 a_1 + x_1 a_2 - x_2 (a_1 + a_2) = (-x_2 + x_1) a_2 \in I_2. \end{aligned}$$

This implies the map is surjective. Its kernel is $I_1 \cap I_2$. Hence

$$R / (I_1 \cap I_2) \cong R / I_1 \oplus R / I_2.$$

To generalize the case of n pairwise relatively prime ideals, we have the map

$R \rightarrow R/I_1 \oplus R/I_2 \oplus \dots \oplus R/I_n$ by sending x to $(x + I_1, x + I_2, \dots, x + I_n)$. Now by induction we assume we find a y such that $y - x_i \in I_i$ for $i = 1, 2, \dots, n-1$. Now since for $1 \leq i \leq n-1$, we have $I_i + I_n = R$, hence $(I_1 \cap I_2 \cap \dots \cap I_{n-1}) + I_n = R$. Now for y and x_n , we are back to the two ideals' case to find a x . And $x - y \in I_1 \cap I_2 \cap \dots \cap I_{n-1}$, hence $x - y \in I_i, 1 \leq i \leq n-1$. Also $x - y \in I_n$. Finally $x - x_i = (x - y) + (y - x_i) \in I_i, 1 \leq i \leq n-1$.

(Note the proof here is not constructive.)

Corollary. $\mathbb{Z} / mn\mathbb{Z} \cong \mathbb{Z} / m\mathbb{Z} \oplus \mathbb{Z} / n\mathbb{Z}$, if $\gcd(m, n) = 1$.

Example. Consider the system

$$\begin{cases} x \equiv 2 \pmod{3}; \\ x \equiv 3 \pmod{4}; \\ x \equiv 1 \pmod{5}. \end{cases}$$

By brute force, or otherwise, we find $x \equiv 11 \pmod{60}$ is the unique solution modulo $3 \times 4 \times 5 = 60$.

Example. Consider the system

$$\begin{cases} 7x \equiv 3 \pmod{12}; \\ 10x \equiv 6 \pmod{14}. \end{cases}$$

Check that the first equation has solutions of the form $x = 9 + 12t$, and the second $x = 2 + 7t$. Hence the system may be changed to $x \equiv 9 \pmod{12}, x \equiv 2 \pmod{7}$. By the Chinese remainder theorem, get the general solution of the form $9 + 84t$.

Example. Consider the system $x^2 \equiv 1 \pmod{3}, x \equiv 2 \pmod{4}$.

The system is equivalent to the two systems $x \equiv 1 \pmod{3}, x \equiv 2 \pmod{4}$; or $x \equiv 2 \pmod{3}, x \equiv 3 \pmod{4}$. Solve to get $x \equiv 10 \pmod{12}$ or $x \equiv 2 \pmod{12}$.

Example. The Euler phi-function $\varphi(m)$ counts the number of integers between 1 and m which are relatively prime to m . It is multiplicative, i.e. $\varphi(mn) = \varphi(m)\varphi(n)$ if $\gcd(m, n) = 1$.

Let j be a number between 0 and $mn-1$. For each of this j takes j_1 be its least nonnegative residue modulo m and j_2 be its least nonnegative residue modulo n . That is to say $j \equiv j_1 \pmod{m}$ with $0 \leq j_1 \leq m-1$ and $j \equiv j_2 \pmod{n}$ with $0 \leq j_2 \leq n-1$. By the Chinese Remainder Theorem, for each pair j_1, j_2 , there is one and only one j between 0 and $mn-1$ with such property. Also j has no common factor with mn if and only if j_1 has no common factor with m and j_2 has no common factor with n , thus counting j is 1-to-1 corresponding to the pairs (j_1, j_2) , and hence $\varphi(mn) = \varphi(m)\varphi(n)$,

For a prime power p^α , by counting those elements which are not relatively prime to p , we have $\varphi(p^\alpha) = p^\alpha - p^{\alpha-1} = p^\alpha(1 - \frac{1}{p})$. Hence if $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_s^{\alpha_s}$, we have

$$\begin{aligned}\varphi(n) &= \varphi(p_1^{\alpha_1})\varphi(p_2^{\alpha_2})\dots\varphi(p_s^{\alpha_s}) \\ &= p_1^{\alpha_1}(1 - \frac{1}{p_1})p_2^{\alpha_2}(1 - \frac{1}{p_2})\dots p_s^{\alpha_s}(1 - \frac{1}{p_s}) = n(1 - \frac{1}{p_1})(1 - \frac{1}{p_2})\dots(1 - \frac{1}{p_s}).\end{aligned}$$

Denote the set of squares modulo i by S_i .

Example: Let $m, n \geq 2$ and such that $\gcd(m, n) = 1$. Then $|S_{mn}| = |S_m||S_n|$.

We consider the map from $\mathbb{Z}/mn\mathbb{Z}$ to $\mathbb{Z}/m\mathbb{Z} \oplus \mathbb{Z}/n\mathbb{Z}$ by sending a to (a, a) . Clearly a is a square modulo mn , then a is a square modulo m and modulo n ($a \equiv x^2 \pmod{mn}$, then $a \equiv x^2 \pmod{m}$ and $a \equiv x^2 \pmod{n}$.) The map is injective since $c \equiv c' \pmod{mn}$ implies $c \equiv c' \pmod{m}$ and $c \equiv c' \pmod{n}$. To show surjectivity, suppose $b \equiv y^2 \pmod{m}$ and $c \equiv z^2 \pmod{n}$, by the Chinese Remainder Theorem, there exists a such that $a \equiv b \pmod{m}$ and $a \equiv c \pmod{n}$. We have to check if a is a square. Indeed by the Theorem again, there exist x such that $x \equiv y \pmod{m}$ and $x \equiv z \pmod{n}$. Hence $x^2 \equiv y^2 \pmod{m}$ and $x^2 \equiv z^2 \pmod{n}$. Thus $a \equiv x^2 \pmod{m}$ and $a \equiv x^2 \pmod{n}$, finally $a \equiv x^2 \pmod{mn}$, hence a is in fact a square. This isomorphism implies $|S_{mn}| = |S_m||S_n|$.

(Check set of squares modulo 3 is $\{0, 1\}$, set of squares modulo 5 is $\{0, 1, 4\}$, and set of squares modulo 15 is $\{0, 1, 4, 6, 9, 10\}$. Another example, set of squares modulo 4 is $\{0, 1\}$, set of squares modulo 9 is $\{0, 1, 4, 7\}$, and finally set of squares modulo 36

is $\{0,1,4,9,13,16,25,28\}$. Note the correspondences.)

Here are more examples.

Example: For every positive integer n , there exists integers a and b such that $4a^2 + 9b^2 - 1$ is divisible by n .

It suffices to find such an a and b modulo a prime power. For 2^k , take $a \equiv 0 \pmod{2^k}$, $b \equiv 3^{-1} \pmod{2^k}$. For any other p^k , take $a \equiv 2^{-1} \pmod{p^k}$ and $b \equiv 0 \pmod{p^k}$.

Example (IMO 1990). Show that for every positive integer n , there exist n consecutive integers such that none of them is a power of a prime.

Use $x+i \equiv 0 \pmod{p_i q_i}$.

Example (Hong Kong Team Selection Test October 2015) Example: 4. Mable and Nora play a game according to the following steps in order.

- (1). Mable writes down any 2015 distinct prime numbers in ascending order in a row. The product of these primes is Mable's score.
- (2). Nora writes down a positive integer.
- (3). Mable draws a vertical line between two adjacent primes she has written in step (1), and compute the product of the prime(s) on the left of the vertical line.
- (4). Nora must add the product obtained by Mable in step (3) to the number she has written in step (2), and the sum becomes Nora's score.

If Mable's and Nora's scores have a common factor greater than 1, Mable wins. Otherwise Nora wins.

Who has a winning strategy?

Nora has a winning strategy. Suppose Mable writes $p_1, p_2, \dots, p_{2015}$ in (1) and let $P = p_1 p_2 \cdots p_{2015}$ be Mable's score. Suppose Nora writes the integer D in (2). Let $q_j = p_1 p_2 \cdots p_j$. Then Nora's score is $D + q_j$ for some $1 \leq j \leq 2014$ at Mable's choice. It suffices to show that there exists D such that $(D + q_j, P) = 1$ for all j , so that Nora must win. For each $1 \leq i \leq 2014$, we choose r_i such that $r_i + q_j$ is not divisible by p_i for all $1 \leq j \leq 2014$. Such r_i must exist, since q_j is divisible by p_i whenever $j \geq i$, so there are at most i 'forbidden values' for r_i (taken modulo p_i)

while we must have $p_i > i$. Now by the Chinese remainder theorem, there exists a positive integer D for which $D \equiv r_i \pmod{p_i}$ for $1 \leq i \leq 2014$. Since $r_i + q_j$ is not divisible by p_i for all $1 \leq j \leq 2014$, $(D + q_j, p_i) = 1$ for all j . Since this is true for all $1 \leq i \leq 2014$, we have $(D + q_j, P) = 1$ for all $1 \leq j \leq 2014$, and the result follows.

Example (Rom MC 2012): A positive integer N is *abundant* if the sum of its factors (1 and N included) exceeds $2N$. Given any positive integer n , there exist n consecutive abundant numbers.

Proof. Let p_k be the k^{th} prime. Recall $\prod_k (1 + \frac{1}{p_k}) = \infty$. Hence for each integer m , we must choose $N(m) \geq m$ such that $\prod_{k=m}^{N(m)} (1 + \frac{1}{p_k}) > 2$. Using this, we check $\prod_{k=m}^{N(m)} p_k$ is abundant.

Set $m_1 = 1$ and $m_{j+1} = N(m_j) + 1, j = 1, 2, 3, \dots$. Set $M_j = \prod_{k=m_j}^{N(m_j)} p_k, j = 1, 2, 3, \dots$. Easy

to see the M_j 's are pairwise relatively prime, so by the Chinese Remainder Theorem, we find M such that $M + j \equiv 0 \pmod{M_j}, j = 1, 2, \dots, n$. Since a multiple of an abundant number is abundant, $M + 1, M + 2, \dots, M + n$ are n consecutive abundant numbers.

Example (Rom MC 2014): Let k be a positive integer and m be a positive odd integer. Show that there exists a positive integer n such that $m^n + n^m$ has at least k distinct prime factors.

We construct a list of k primes $p_1 < p_2 < \dots < p_k$ as follows. First $p_1 > 2m$. After selecting p_j we use Dirichlet's theorem to choose p_{j+1} as

$$p_{j+1} \equiv -1 \pmod{p_1(p_1-1)p_2(p_2-1)\dots p_j(p_j-1)}.$$

If $i < j$, then $p_i < p_j$, so $p_j \nmid (p_i - 1)$. Moreover $p_j - 1 \equiv -2 \pmod{p_i(p_i-1)}$, so $p_i \nmid (p_j - 1)$.

We now use the Chinese Remainder Theorem to choose a positive integer n such that $n \equiv -1 \pmod{p_1 p_2 \dots p_k}$ and $n \equiv 0 \pmod{(p_1 - 1)(p_2 - 1) \dots (p_k - 1)}$.

Finally since m and $p_1 p_2 \dots p_k$ are relatively prime, $(p_1 - 1)(p_2 - 1) \dots (p_k - 1) \mid n$, and m is odd, by Euler's Theorem, we have

$$m^n + n^m \equiv 1 + (-1)^m \equiv 0 \pmod{p_1 p_2 \dots p_k}, \text{ and the result follows.}$$

Example: Let $n > 1$ be an odd integer; let k be an integer such that for every prime

factor p of n , $(p - 1) \mid k$. Let $S_k(n) = \sum_{\substack{i=1 \\ \gcd(i, n)=1}}^n i^k$. Then $n \mid S_k(n)$.

(For examples, $n = 5, k = 3$, get $S_3(5) = 1^3 + 2^3 + 3^3 + 4^3 = 100$ is divisible by 5;

$n = 15, k = 3$, get $S_3(15) = 1^3 + 2^3 + 4^3 + 7^3 + 8^3 + 11^3 + 13^3 + 14^3 = 7200$ is divisible by 15, note we cannot take $k = 2$.)

First we try $n = p$ an odd prime, if a is relatively prime to p , we have

$$S_k(p) = 1^k + 2^k + \dots + (p - 1)^k \equiv (a1)^k + (a2)^k + \dots + (a(p - 1))^k = a^k S_k(p), \text{ since } a \text{ just}$$

permutes the residue classes $1, 2, \dots, (p - 1)$. Thus we have $(a^k - 1)S_k(p) \equiv 0 \pmod{p}$.

Now if we can choose a such that $p \nmid (a^k - 1)$ and as $\gcd(a, p) = 1$, then we will have $S_k(p) \equiv 0 \pmod{p}$, and we are done. Indeed since $(p - 1) \mid k$ we have $k = (p - 1)q + r$ with $0 < r \leq p - 2$, the equation $x^r - 1 \equiv 0 \pmod{p}$ has at most r solutions we can find at least one member from $1, 2, \dots, (p - 1)$ such that $p \nmid (a^r - 1)$ and we are done.

Now n is an odd number and a is relatively prime to n . Analogously we can prove

$$(a^k - 1)S_k(n) \equiv 0 \pmod{n}. \text{ If we can find } a \text{ such that } \gcd(a, n) = \gcd(a^k - 1, n) = 1,$$

then we are done. Indeed we have the prime factors of n , $p_1, p_2, \dots, p_l$, and for each

prime p_i , we can find a_{p_i} such that $\gcd(a_{p_i}, p_i) = \gcd(a_{p_i}^k - 1, p_i) = 1$. Using the

Chinese Remainder Theorem, we can find a satisfying the l congruence relations

$$a \equiv a_{p_i} \pmod{p_i}, 1 \leq i \leq l. \text{ Clearly } \gcd(p_i, a) = 1, \text{ hence } \gcd(a, n) = 1. \text{ Also}$$

$$a^k - 1 \equiv a_{p_i}^k - 1 \pmod{p_i}, \text{ hence } p_i \nmid (a^k - 1), \text{ thus } \gcd(a^k - 1, n) = 1 \text{ and we are done.}$$

Example (RMC 2012) Determine all positive integers n satisfying the following condition: the sum of squares of n different prime numbers greater than 3 is divisible by n .

Let n is such a number and k is relatively prime to n , then $k^2 \equiv 1 \pmod{n}$. Indeed we can choose $n-1$ different primes congruent to 1 mod n and a prime congruent to $k \pmod{n}$, using Dirichlet. Then the sum of squares of these primes is congruent to $k^2 + (n-1) \equiv k^2 - 1 \pmod{n}$. Hence $k^2 \equiv 1 \pmod{n}$.

Now we want to prove n has no prime divisor greater than 3. Indeed let p be an odd divisor of n and $n = p^\alpha m$, where p does not divide m . By the Chinese Remainder Theorem, there exists a positive integer k such that $k \equiv 1 \pmod{m}$ and $k \equiv 2 \pmod{p}$. Clearly such a k is relatively prime to n , so $k^2 \equiv 1 \pmod{n}$, hence $k^2 \equiv 1 \pmod{p}$, and the condition $k \equiv 2 \pmod{p}$ implies $p \equiv 3$.

Thus n is of the form $n = 2^\alpha 3^\beta$. As 5 is relatively prime to n we have $5^2 \equiv 1 \pmod{2^\alpha 3^\beta}$, we get n a divisor of 24 and all of them work.

Example (USAMO 1982, 4) Find a positive integer k such that $k2^n + 1$ is composite for all positive integer n .

Choose k so that $k > 1$, $k \equiv 1 \pmod{2^{2^h} + 1}$, $h = 0, 1, 2, 3, 4$. It is possible since the Fermat numbers F_0, F_1, F_2, F_3, F_4 are primes. Then if n is of the form $2^h q$, where q is odd and $0 \leq h \leq 4$, we have

$$k \cdot 2^n + 1 = k \cdot 2^{2^h q} + 1 \equiv (1)(2^{2^h})^q + 1 \equiv (1)(-1)^q + 1 \equiv (-1) + 1 \equiv 0 \pmod{2^{2^h} + 1}. \quad \text{Hence}$$

the numbers are composite. Now what happen if $n = 2^h q$, $h \geq 5$? Indeed these numbers are of the form $n = 2^h q = 2^5 m$, where m can be even or odd. We note

$F_5 = 2^{2^5} + 1 = (641) \times (6700417)$, where $P = 641$ and $Q = 6700417$ are distinct primes. We can on the same time, choose k such that $k > 1$, $k \equiv -1 \pmod{P}$ and $k \equiv 1 \pmod{Q}$. Then

$$k2^n + 1 = k2^{2^5 m} + 1 \equiv (-1)(2^{2^5})^m + 1 \equiv (-1)(-1)^m + 1 \equiv (-1)^{m+1} + 1 \pmod{P}.$$

On the other hand, $k2^n + 1 = k2^{2^5 m} + 1 \equiv (1)(2^{2^5})^m + 1 \equiv (-1)^m + 1 \pmod{Q}$.

If m is even, then $k2^n + 1$ is a multiple of P . If m is odd, then $k2^n + 1$ is a multiple

of Q . Hence they are composite. To conclude, choose $k \equiv 1 \pmod{M}$, where $M = 3, 5, 17, 257, 65537, 6700417$, and $k \equiv -1 \pmod{641}$.

Example: (Mongolian MO 2014) Can you move a needle of length l , initially at a vertical position, to a horizontal position by passing through no more than 2015 lattice points?

Yes, the Chinese Remainder Theorem implies the system

$$\begin{cases} n \equiv p_0 \pmod{p_0^2}; \\ n+1 \equiv p_1 \pmod{p_1^2}; \\ \dots \\ n+k \equiv p_k \pmod{p_k^2}. \end{cases}$$

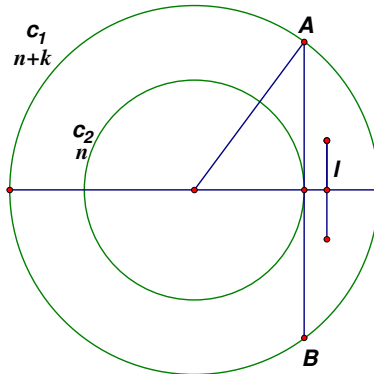
has a positive integer solution for all k , where $p_0, p_1, \dots, p_k$ are primes of the form $4t+3$.

From this we can show there is no lattice points in the ring $\{(x, y) \in \mathbb{R}^2 \mid n \leq x^2 + y^2 \leq n+k\}$. Indeed if so, then $x^2 + y^2 = n+t$, $0 \leq t \leq k$, $x, y \in \mathbb{Z}$.

Thus $x^2 + y^2 \equiv p_t \pmod{p_t^2}$ and $x^2 + y^2 \equiv 0 \pmod{p_t}$. But $x^2 + y^2 \not\equiv 0 \pmod{p_t^2}$ and

$p_t \equiv 3 \pmod{4}$ imply $x \equiv y \equiv 0 \pmod{p_t}$ since otherwise $\left(\frac{x}{y}\right)^2 \equiv -1$ a square.

Hence $p_t^2 \mid x^2 + y^2$, which is a contradiction. Thus there is no lattice point in the ring.



From the diagram, the inner circle is of radius $\sqrt{n}$ and outer circle $\sqrt{n+k}$, hence AB is of length $2\sqrt{k}$. If it is larger than l (or $k^2 > \frac{l^2}{4}$), then we can move the needle around the ring without encountering any lattice point, to a horizontal position.

Exercises

1. (Vietnam MO 2015) A positive integer k has the property $T(m)$ if for any positive integer a , there exists a positive integer n such that $1^k + 2^k + \dots + n^k \equiv a \pmod{m}$.

(a) Find all positive integer k with the property $T(20)$.

(b) Find the smallest positive integer k with the property $T(20^{15})$.

2. Let n be a positive integer. Determine in terms of n , the number of x from 1 to n such that $x^2 \equiv x \pmod{n}$.

3. (IMO2009) Let n be a positive integer and $a_1, a_2, \dots, a_k, k \geq 2$, be distinct integers in the set $\{1, 2, \dots, n\}$ such that $n \mid a_i(a_{i+1} - 1)$ for $i = 1, 2, \dots, k-1$. Show that $n \nmid a_k(a_1 - 1)$.

4. (APMO2009) For any positive integer k , there exists an arithmetic progression

$\frac{a_1}{b_1}, \frac{a_2}{b_2}, \dots, \frac{a_k}{b_k}$ of rational numbers, where a_i, b_i for all $i = 1, 2, \dots, k$, are relatively prime positive integers and moreover the $2k$ numbers $a_1, \dots, a_k, b_1, \dots, b_k$ are pairwise distinct.

5. (ELMO Shortlist) Find all triples (a, b, c) of positive integers such that if n is a positive integer not divisible by any prime less than 2014, then $n+c$ divides $a^n + b^n + n$.

6. Let $a > b > c \geq 3$ be integers. Given that $a \mid bc + b + c$, $b \mid ca + c + a$ and $c \mid ab + a + b$, prove that at least one of a, b, c is not prime.

7. Prove that for every positive integer n , there exists a set S of n positive integers such that for any two distinct $a, b \in S$, $a-b$ divides a and b but none of the other

elements of S .

8. For each positive integer n , there are pairwise relatively prime integers $k_0, k_1, \dots, k_n$, are strictly greater than 1, such that $k_0 k_1 \dots k_n - 1$ is the product of two consecutive integers.

9. Let N be the set of positive integers. Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a function satisfying the following two conditions:

- (a) $f(m)$ and $f(n)$ are relatively prime whenever m and n are relatively prime.
- (b) $n \leq f(n) \leq n + 2012$ for all n .

Prove that for any natural number n and any prime p , if p divides $f(n)$, then p divides n .

10. (USAMO2014): Prove that there exists a constant $c > 0$ with the following property: If a, b, n are positive integers such that $\gcd(a+i, b+j) > 1$ for all $i, j \in \{0, 1, \dots, n\}$, then $\min\{a, b\} > (cn)^n$.

References

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Hints to Exercises

1. (a) Note $n^{k+4} - n^k$ is divisible by 20 so we just have to check the cases for $n=1,2,3,4,5$. Now 1, 2, 3, 5 are no good, $k=4$ is OK, so k has $T(20)$ if and only if k is divisible by 4. (b) $k=4$ has the property $T(20^{15})$. Let

$S_4(n) = 1^4 + 2^4 + \dots + n^4$, and $S(n) = 30S_4(n)$. We have to show

$S(n) \equiv 30a \pmod{30 \times 20^{15}}$ is solvable for any a . This is reduced to solving the system $S(n) \equiv 30a \pmod{3}$, $S(n) \equiv 30a \pmod{2^{31}}$, $S(n) \equiv 30a \pmod{5^{16}}$, by the Chinese Remainder Theorem. The last two equations can be done by induction.

2. Solve the problem for prime powers dividing n . Answer is a power of 2.

3. By contradiction. Look for a prime power $p^r | n$, and show that either

$a_i \equiv 0 \pmod{p^r}$ or $a_i \equiv 1 \pmod{p^r}$, for all i .

4. Pick x, N such that $\frac{x+1}{N}, \dots, \frac{x+k}{N}$ works after reducing to the lowest terms.

Force exactly one prime to cancel in each reduction.

5. Pick n modulo p and $p-1$ for a large prime p . Force p to divide a constant.

6. No comment.

7. For $n=3$, $x_1 \underset{2}{\wr} x_2 \underset{3}{\wr} x_3$ gives $\{10, 12, 15\}$. For $n=4$, $x_1 \underset{60}{\wr} x_2 \underset{90}{\wr} x_3 \underset{7}{\wr} x_4$.

8. We have to show $t(t+1)+1 = k_0 k_1 \dots k_n$ may have arbitrarily many prime factors. It suffices to show that for any n there exist a t such that $P(t) = t(t+1) = t^2 + t + 1$ has more than $n+1$ prime factors.

So we have to construct a t such that $P(t) \equiv 0 \pmod{p_i}$ for some $n+1$ primes. For this we have to find t_i such that $P(t_i) \equiv 0 \pmod{p_i}$ for each i , and then apply the Chinese Remainder Theorem to find a t such that $t \equiv t_i \pmod{p_i}$, meaning $P(t) \equiv P(t_i) \equiv 0 \pmod{p_i}$.

Finally we just have to show there are infinitely many primes dividing some number of the form $t(t+1)$. Indeed if P is the set of all primes dividing numbers of the form $t^2 + t + 1$, we take N the product of all primes in P and consider $N^2 + N + 1$ and get a

prime not in the list P , dividing some number of the form $t^2 + t + 1$, a contradiction.

9. Show that for any natural number n , and for any prime p , if p divides $f(n)$, then p divides n .

(Idea) Fix n and p , assume by contradiction $p \nmid n$.

Now the idea is we want to find a large number N such that $f(N) = N$. We first pick 2012×2013 distinct primes $q_{i,j} > n + p + 2013$ for every $i = 1, 2, \dots, 2012$ and $j = 0, 1, 2, \dots, 2012$; we then use the primes to fill in the table:

	$N + 1$	$N + 2$	...	$N + 2012$
M	$q_{0,1}$	$q_{0,2}$	...	$q_{0,2012}$
$M + 1$	$q_{1,1}$	$q_{1,2}$	...	$q_{1,2012}$
...	...	...	...	...
$M + 2013$	$q_{2012,1}$	$q_{2012,2}$	...	$q_{2012,2012}$

By the Chinese Remainder Theorem, we can find N such that $N + 1 \equiv 0 \pmod{q_{i,1}}$; and

similarly for $N + 2$, etc. We add further condition $N \equiv 0 \pmod{p}$ and $N \equiv 1 \pmod{n}$.

Note that N is not divisible by any $q_{i,j}$'s since they are larger than 2012.

We choose M accordingly with further conditions $M \equiv 1 \pmod{N}$.

Now $M \equiv 1 \pmod{N}$, hence $f(M)$ and $f(N)$ are relatively prime. But from the table $f(M)$ must have a common factor with each of $N + 1, N + 2, \dots, N + 2012$.

Thus the only possibility is $f(N) = N$.

Now $N \equiv 1 \pmod{n}$, hence $\gcd(N, n) = 1$, thus

$1 = \gcd(f(N), f(n)) = \gcd(N, f(n))$. But $p \mid N$ and $p \mid f(n)$, a contradiction.

10. For some $0 \leq i, j \leq n$, some primes divide both $a + i$ and $b + j$. Write down a $(n + 1) \times (n + 1)$ table so that the (i, j) position is occupied by a prime (as small as possible) that divide both $a + i$ and $b + j$. We know these primes are well spaced and “sparse”. Hence we run out of small primes soon.

With enough calculations with $\sum_p \left\lceil \frac{n+1}{p} \right\rceil^2$, we can show half of the table consists of

primes greater than $0.001n^2$. Hence for some column at least half of its entries consist of primes larger than $0.001n^2$. Thus $a+i$ is divisible by product of primes greater than $(0.001n^2)^{n/2} = (cn)^n$, and we are done.

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